NAME: _____ Geometry (FL B.E.S.T.)

11.1 Copying Segments and Angles

Classical constructions use only two tools: an **unmarked straightedge** (draws lines, but never measures) and a **compass** (transfers a fixed distance). Every construction is a sequence of steps that can each be *justified* — usually by showing that the construction forces two triangles to be congruent (often by SSS). This worksheet builds the two most fundamental constructions — copying a segment and copying an angle — and uses them to copy an entire triangle. Each example gives a full, justified step list, not just a final picture: on a real exam you must be able to explain *why* each step guarantees congruence, not just perform it.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. What is the first step to copy $\overline{MN}$ onto a ray from M'?

2. Where do you place the compass point and pencil to set width MN?

3. After arcing at the original vertex, what is drawn next at the new vertex?

4. What length is transferred to place the second ray when copying an angle?

5. In an SSS triangle copy, after copying side AB as DE, which two lengths locate point F?

6. A copied angle is congruent to the original because of which triangle congruence postulate?

7. Why is a protractor not allowed in a classical angle copy?

8. Can a triangle be constructed with sides 3, 4, and 9? Justify using the Triangle Inequality.

9. After copying $\overline{AB}$ to $\overline{A'B'}$, what relationship holds, and why?

10. Why must the straightedge used in constructions be unmarked?

11. How does copying a segment differ from bisecting one?

12. In an angle copy, what do the two points where the first arc crosses the rays represent?

13. What does completing the SSS triangle construction guarantee about $\triangle DEF$ and $\triangle ABC$?

14. An angle $(5x - 4)^\circ$ is copied exactly, and the copy measures 41°. Find x.

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15. You are given three segments of lengths 5, 12, and 13 to build $\triangle DEF \cong \triangle ABC$ by SSS. (a) Verify a triangle is possible using the Triangle Inequality. (b) Explain why SSS produces exactly one triangle shape (up to flipping it over), not several different ones.

✓ Answer Key

1. Draw a ray starting at M'
2. Point on M, pencil on N
3. The same-radius arc, crossing the new ray
4. The chord distance between the two arc–ray intersection points
5. AC (from D) and BC (from E)
6. SSS
7. Constructions must rely on compass-transferred congruence, not measured degree readings
8. No — $3 + 4 = 7 < 9$
9. $\overline{A'B'} \cong \overline{AB}$, because the compass radius never changed
10. So that all lengths come from compass transfers, not from reading numbers off a ruler
11. Copying reproduces the same length elsewhere; bisecting splits one segment into two equal halves
12. Two points equidistant from the vertex, used to build the congruent triangle for SSS
13. $\triangle DEF \cong \triangle ABC$
14. $x = 9$
15. (a) $5 + 12 = 17 > 13$, $5 + 13 > 12$, $12 + 13 > 5$ — valid triangle. (b) Fixing all three side lengths leaves no freedom to change any angle, so the triangle's shape is rigid and unique up to reflection